

Mathematics

Topic: Fractions, Percentages, and Financial Mathematics

Scenario 1

In **Luweero District**, a farmers' group harvested 1,200 bags of maize. The leaders decided to sell the harvest in three phases to maximize profit:

- In the first phase, they sold 31 of the total bags at UGX 80,000 per bag.
- In the second phase, they sold 60% of the **remaining** bags at a higher price of UGX 100,000 per bag because of market demand.
- In the final phase, the remaining bags were sold at a discount price of UGX 70,000 per bag before they spoiled.

Transporting the maize to the market cost UGX 2,500,000 in total, and storage fees cost UGX 1,500,000. The chairperson wants to know the exact net profit made by the group and how much money each of the 40 members should receive if profits are shared equally.

Task

- **Part (a):** Write a detailed explanation showing step-by-step mathematical calculations to find the total revenue earned from all three phases and the net profit after subtracting all costs.
- **Part (b):** Write an analytical report for the chairperson explaining how the profit should be distributed among the members, and explain how selling in phases affected the overall average selling price per bag compared to selling all bags at once for UGX 85,000.

ITEM 2: S3 Mathematics

Topic: Coordinate Geometry, Area, and Optimization (Linear Programming Concepts)

Scenario

A local youth cooperative in **Jinja** received a grant to fence off a triangular plot of land near the riverbank for a community vegetable garden. The survey markers for the plot boundary are recorded on a grid map with coordinates measured in meters: **Point A (2,3)**, **Point B (10,3)**, and **Point C (6,11)**.

The cooperative needs to buy chain-link fencing wire for the entire perimeter and organic fertilizer for the interior planting area. One roll of fencing wire covers 10 meters and costs UGX 150,000 (partial rolls cannot be bought). One bag of organic fertilizer covers 12 square meters of soil and costs UGX 25,000.

Task

- **Part (a):** Write an explanation using distance and area formulas to calculate the exact perimeter and area of the triangular plot on the grid map.
- **Part (b):** Write a practical budget report for the cooperative showing the exact number of fencing rolls and fertilizer bags needed, calculating the total cost, and explaining mathematically why rounding up full units is necessary for real-life purchases.